Lecture 20: Chebychev Nodes; Spline Spaces and B-Splines

Math 6610

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Today's Goals

- Finish Lecture 19: interpolation error (with proofs); error of splines
- Splines $\mathcal{S}_{k,\Delta}$ as a vector space of dimension $k+n-1$
- Define B-splines N_{ik} by the Cox–de Boor recursion

Interpolation Error: Statements

Theorem (*Interpolation Error I*). Let $f^{(n+1)}$ be continuous on $[a,b]$, and let $p_n \in \prod_n$ interpolate f at distinct nodes $x_0, \ldots, x_n \in [a,b]$. Then for every $x \in [a,b]$ there is $c = c(x) \in (a,b)$ with

$$e(x) := f(x) - p_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!} \prod_{i=0}^{n} (x - x_i).$$

Lemma (*Bounding Lemma*). Let $n \geqslant 1$, $h = (b-a)/n$ and $x_i = a + ih$, $i = 0, \ldots, n$. Then for every $x \in [a,b]$

$$\prod_{i=0}^{n} |x - x_i| \leqslant \frac{h^{n+1}}{4}\, n!.$$

Remark. The theorem holds for any distinct nodes; the lemma is for equally spaced nodes only.

Interpolation Error I: Proof

Proof. Fix $x \in [a,b]$. Let $\omega(t) := \prod_{i=0}^{n}(t-x_i)$ and $e(t) := f(t) - p_n(t)$.
1. x is a node, $x = x_q$: $e(x) = 0$ and $\omega(x) = 0$, so the formula holds with any $c \in (a,b)$.
2. x is not a node: then $\omega(x) \neq 0$. Define (K, g depend on x; so will c)

$$K := \frac{e(x)}{\omega(x)}, \qquad g(t) := e(t) - K\,\omega(t), \quad t \in [a,b].$$

3. Smoothness. $f \in C^{n+1}[a,b]$; p_n, ω are polynomials. Hence $g \in C^{n+1}[a,b]$.
4. Zeros of g. At the nodes and at x:

$$g(x_i) = e(x_i) - K\,\omega(x_i) = 0 - K \cdot 0 = 0, \qquad i = 0, \ldots, n,$$

$$g(x) = e(x) - \frac{e(x)}{\omega(x)}\,\omega(x) = 0.$$

So g has $n+2$ distinct zeros $s_0 < \cdots < s_{n+1}$ in $[a,b]$.

(continued on the next slide)

Interpolation Error I: Proof (cont'd)

Proof (continued). 5. Rolle, $n+1$ times. If $g^{(m)}$, $m \leqslant n$, has zeros $s_0 < \cdots < s_{n+1-m}$, Rolle on each $[s_r, s_{r+1}]$ gives

$$g^{(m+1)}(s_r') = 0, \quad s_r' \in (s_r, s_{r+1}), \quad r = 0, \ldots, n-m:$$

$n+1-m$ distinct zeros (disjoint intervals). Induction on m from step 4:

$$\text{distinct zeros:} \quad g: \ n+2, \quad g': \ n+1, \quad \ldots, \quad g^{(n+1)}: \ 1.$$

So $g^{(n+1)}(c) = 0$ for some $c \in (s_0, s_{n+1}) \subseteq (a, b)$.
6. Differentiate $n+1$ times. $p_n \in \prod_n$ and $\omega(t) = t^{n+1} + (\text{degree} \leqslant n)$, so

$$p_n^{(n+1)} \equiv 0, \qquad \omega^{(n+1)} \equiv (n+1)!, \qquad g^{(n+1)}(t) = f^{(n+1)}(t) - K\,(n+1)!.$$

7. Evaluate at c and solve for $e(x)$:

$$0 = f^{(n+1)}(c) - K\,(n+1)! \implies K = \frac{f^{(n+1)}(c)}{(n+1)!} \implies e(x) = K\,\omega(x) = \frac{f^{(n+1)}(c)}{(n+1)!}\,\omega(x).$$

Bounding Lemma: Proof

Proof. 1. Locate $x \in [a, b]$. Choose $j \in \{0, \ldots, n-1\}$ with $x \in [x_j, x_{j+1}]$ ($j = n-1$ if $x = b$); write $x = x_j + s\,h$, $s \in [0, 1]$.
2. The two nearest nodes x_j, x_{j+1}:

$$\left|x - x_j\right| \left|x - x_{j+1}\right| = sh \cdot (1-s)h = s(1-s)\,h^2 = \left(\tfrac{1}{4} - \left(s - \tfrac{1}{2}\right)^2\right)h^2 \leqslant \frac{h^2}{4}.$$

3. Nodes to the left, $i = 0, \ldots, j-1$: $|x - x_i| = (j - i + s)h \leqslant (j + 1 - i)h$, so

$$\prod_{i=0}^{j-1} |x - x_i| \leqslant \left[(j+1) \cdot j \cdots 2\right] h^j = (j+1)!\,h^j.$$

4. Nodes to the right, $i = j+2, \ldots, n$: $|x - x_i| = (i - j - s)h \leqslant (i - j)h$, so

$$\prod_{i=j+2}^{n} |x - x_i| \leqslant \left[2 \cdot 3 \cdots (n-j)\right] h^{n-j-1} = (n-j)!\,h^{n-j-1}.$$

(continued on the next slide)

Bounding Lemma: Proof (cont'd)

Proof (continued). 5. Multiply steps 2, 3, 4 (empty products $= 1$; powers of h: $2 + j + (n - j - 1) = n + 1$):

$$\prod_{i=0}^{n} |x - x_i| \leqslant \frac{h^2}{4} \cdot (j+1)!\, h^j \cdot (n-j)!\, h^{n-j-1} = \frac{h^{n+1}}{4}\, (j+1)!\, (n-j)!.$$

6. Write the factorials as a binomial coefficient, $q := j + 1$:

$$(j+1)!\, (n-j)! = q!\, (n+1-q)! = \frac{(n+1)!}{\binom{n+1}{q}}.$$

7. Bound it. $1 \leqslant q \leqslant n$; the row $\binom{n+1}{q}$ rises to the middle and is symmetric, so $\binom{n+1}{q} \geqslant \binom{n+1}{1} = n + 1$. Insert into steps 6 and 5:

$$(j+1)!\, (n-j)! \leqslant \frac{(n+1)!}{n+1} = n!, \qquad \prod_{i=0}^{n} |x - x_i| \leqslant \frac{h^{n+1}}{4}\, n!.$$

Interpolation Error II

Corollary (*Interpolation Error II*). With f, p_n as in the theorem, nodes as in the lemma, and $M := \max_{[a,b]} \left|f^{(n+1)}\right|$: $\max_{[a,b]} |e(x)| \leqslant \dfrac{Mh^{n+1}}{4(n+1)}$.

Proof. *1. M exists:* $f^{(n+1)}$ is continuous on the closed interval.
2. Fix $x \in [a,b]$; c from Interpolation Error I:

$$
\begin{aligned}
|e(x)| &= \frac{\left|f^{(n+1)}(c)\right|}{(n+1)!} \prod_{i=0}^{n} |x - x_i| && \text{(Interpolation Error I)} \\
&\leqslant \frac{M}{(n+1)!} \cdot \frac{h^{n+1}}{4}\, n! && \left(\left|f^{(n+1)}(c)\right| \leqslant M\text{; lemma}\right) \\
&= \frac{Mh^{n+1}}{4(n+1)} && \left((n+1)! = (n+1)\, n!\right).
\end{aligned}
$$

3. The bound is independent of x: take the maximum over x. ∎

Remark. M depends on n; if it grows fast, the bound need not tend to 0 (Runge's example below).

Understanding the Error Formula

$$e(x) = \underbrace{\frac{f^{(n+1)}(c)}{(n+1)!}}_{\text{how smooth } f \text{ is}} \underbrace{\omega(x)}_{\text{where the nodes are}}, \qquad \omega(x) = \prod_{i=0}^{n}(x - x_i).$$

- Bound for any nodes: with $M = \max_{[a,b]} \left|f^{(n+1)}\right|$,

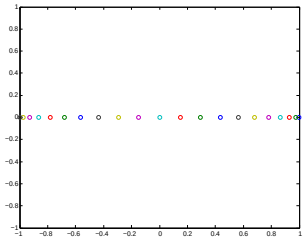
$$\max_{[a,b]} |e| \leqslant \frac{M}{(n+1)!} \max_{x\in[a,b]} |\omega(x)| .$$

- M is fixed by f. Only ω is ours to choose, through the nodes.
- Equally spaced, $n = 10$ on $[-1, 1]$: $\max|\omega| \approx 8.5 \cdot 10^{-3}$, reached in the first and last intervals (picture: “The Node Product”).
- Question: which nodes in $[-1, 1]$ make $\max_{[-1,1]} |\omega|$ as small as possible?

Chebychev Nodes

Definition (*Chebychev nodes*). The Chebychev nodes of degree n in $[-1,1]$ are

$$x_i = \cos\left(\pi \frac{2i+1}{2n+2}\right), \qquad i = 0, \ldots, n; \qquad \text{on } [a,b]: \ \tfrac{a+b}{2} + \tfrac{b-a}{2} x_i.$$



- Equally spaced angles on the half circle, projected down to the x-axis
- Not uniform (bunched at the ends) and not nested (new n, new nodes)

Chebychev Polynomials

Definition (*Chebychev polynomial*). For $m = 0, 1, 2, \ldots$ and $x \in [-1, 1]$, $T_m(x) := \cos(m \arccos x)$. Equivalently, with $x = \cos\theta$, $\theta \in [0, \pi]$: $T_m(\cos\theta) = \cos(m\theta)$.

Lemma. $T_0(x) = 1$, $T_1(x) = x$, and $T_{m+1}(x) = 2x\,T_m(x) - T_{m-1}(x)$ for $m \geqslant 1$. Hence, for $m \geqslant 1$, T_m is a polynomial of degree m with leading coefficient 2^{m-1}.

Proof. 1. Recurrence. The identity $\cos((m+1)\theta) + \cos((m-1)\theta) = 2\cos\theta\,\cos(m\theta)$ with $x = \cos\theta$ reads $T_{m+1} + T_{m-1} = 2x\,T_m$.
2. Degree, induction on m. Base: $T_1 = x$ has degree 1 and leading coefficient $1 = 2^0$. Step: if T_m has degree m, leading coefficient 2^{m-1}, and T_{m-1} has degree $m-1$, then $2x\,T_m - T_{m-1}$ has degree $m+1$ and leading coefficient $2 \cdot 2^{m-1} = 2^m$. ■

Example. $T_2 = 2x^2 - 1$, $T_3 = 4x^3 - 3x$. (Check: $2x(2x^2 - 1) - x = 4x^3 - 3x$.)

Zeros and Extremes of T_{n+1}

Proposition. Let $n \geqslant 0$.

(i) The zeros of T_{n+1} are exactly the $n+1$ Chebychev nodes $x_i = \cos\left(\pi \frac{2i+1}{2n+2}\right)$, $i = 0, \ldots, n$.

(ii) $|T_{n+1}(x)| \leqslant 1$ for $x \in [-1, 1]$, and at the $n+2$ points $y_j := \cos \frac{j\pi}{n+1}$, $j = 0, \ldots, n+1$, $T_{n+1}(y_j) = (-1)^j$.

Proof. Write $x = \cos\theta$; this is one-to-one from $\theta \in [0, \pi]$ onto $x \in [-1, 1]$.
(i) $\cos((n+1)\theta) = 0 \Leftrightarrow (n+1)\theta = (2i+1)\frac{\pi}{2}$; $\theta \in [0, \pi]$ allows exactly $i = 0, \ldots, n$. These $n+1$ zeros are distinct, and a polynomial of degree $n+1$ has no more.
(ii) $|\cos| \leqslant 1$, and $\theta_j = \frac{j\pi}{n+1}$ gives $\cos((n+1)\theta_j) = \cos(j\pi) = (-1)^j$. ∎

Remark. $1 = y_0 > y_1 > \cdots > y_{n+1} = -1$: T_{n+1} swings between $+1$ and -1 exactly $n+2$ times. This alternation is the key to the next proof.

Chebychev Nodes Are Optimal

Theorem. Let $x_0, \ldots, x_n$ be the Chebychev nodes and $\omega(x) = \prod_{i=0}^{n}(x - x_i)$.

(a) $\omega = 2^{-n}T_{n+1}$; hence $\max_{[-1,1]} |\omega| = 2^{-n}$.

(b) For any $z_0, \ldots, z_n \in \mathbb{R}$, $\max_{x\in[-1,1]} \prod_{i=0}^{n} |x - z_i| \geqslant 2^{-n}$.

Proof. 1. Part (a). ω and $2^{-n}T_{n+1}$ both have degree $n+1$, leading coefficient 1 ($2^{-n} \cdot 2^n$), and the same $n+1$ zeros (Proposition (i)). Their difference has degree $\leqslant n$ and $n+1$ zeros, so it is 0. By (ii), $\max |T_{n+1}| = 1$ (reached at $y_0 = 1$).

2. Part (b): assume the opposite. Let $q(x) := \prod_i (x - z_i)$ with $|q(x)| < 2^{-n}$ for all $x \in [-1, 1]$, and set

$$r := 2^{-n}T_{n+1} - q.$$

3. Degree. Both terms are $x^{n+1} +$ (lower powers); the x^{n+1} cancels: $r \in \prod_n$.

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Chebychev Nodes Are Optimal (cont'd)

Proof (continued). 4. Signs. At the points y_j of (ii), $j = 0, \ldots, n+1$,

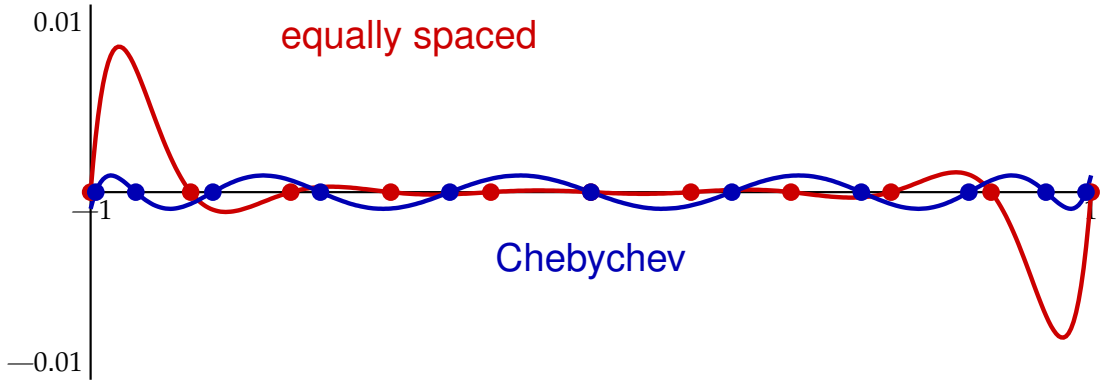
$$r(y_j) = (-1)^j 2^{-n} - q(y_j), \qquad \left|q(y_j)\right| < 2^{-n},$$

so $r(y_j)$ has the sign of $(-1)^j$: r alternates in sign at $n+2$ points.
5. Zeros. r is continuous; by the intermediate value theorem it has a zero in each of the $n+1$ disjoint intervals (y_{j+1}, y_j), $j = 0, \ldots, n$.
6. Contradiction. $r \in \prod_n$ with $n+1$ distinct zeros, so $r \equiv 0$. But $r(y_0) > 0$. ■

Remark. In words: no choice of $n+1$ nodes makes the factor $\max|\omega|$ in the error bound smaller than the Chebychev nodes do. The proof used only the alternation of T_{n+1} at $n+2$ points.

The Node Product: A Picture

Example. $\omega(x) = \prod_i (x - x_i)$ for $n = 10$:



- Equally spaced: $\max|\omega| \approx 8.5 \cdot 10^{-3}$, large near the ends.
- Chebychev: $\max|\omega| = 2^{-10} \approx 9.8 \cdot 10^{-4}$; $|\omega| = 2^{-10}$ at all 12 points y_j (equal ripples).

Error Bound with Chebychev Nodes

Corollary. Let $f^{(n+1)}$ be continuous on $[a,b]$, $M := \max_{[a,b]} \left|f^{(n+1)}\right|$, and let p_n interpolate f at the Chebychev nodes on $[a,b]$, $\hat{x}_i = \frac{a+b}{2} + \frac{b-a}{2} x_i$. Then

$$\max_{[a,b]} |f - p_n| \leqslant \frac{M}{(n+1)!} \cdot 2\Big(\frac{b-a}{4}\Big)^{n+1}.$$

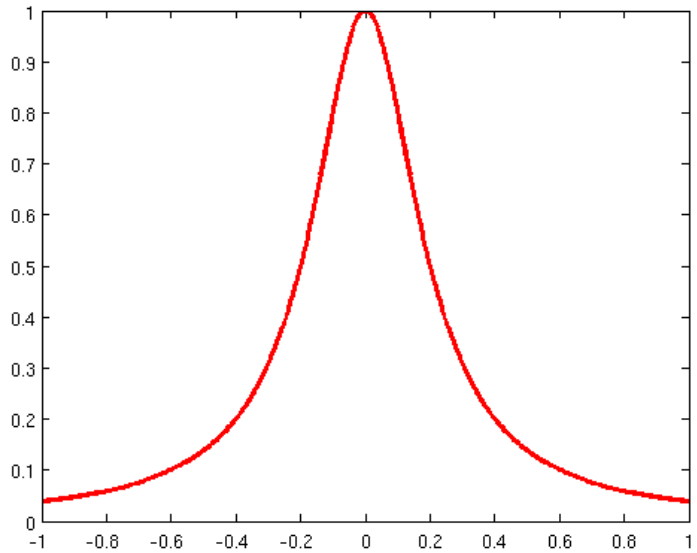
Proof. 1. Change of variable. Each $x \in [a,b]$ is $x = \frac{a+b}{2} + \frac{b-a}{2} t$ with $t \in [-1,1]$; then $x - \hat{x}_i = \frac{b-a}{2}(t - x_i)$.
2. Product. By the theorem, part (a),

$$\prod_{i=0}^{n} |x - \hat{x}_i| = \Big(\frac{b-a}{2}\Big)^{n+1} \prod_{i=0}^{n} |t - x_i| \leqslant \Big(\frac{b-a}{2}\Big)^{n+1} 2^{-n} = 2\Big(\frac{b-a}{4}\Big)^{n+1}.$$

3. Insert into the bound of "Understanding the Error Formula". ■

Equally Spaced vs. Chebychev Nodes

High degree polynomials using equispaced points suffer from many oscillations



- Chebychev nodes: the error is distributed more evenly

Runge's Example: The Experiment

Interpolate $f(x) = 1/(1+25x^2)$ with degree n; measure the maximum error on a fine grid:

```
f  = @(x) 1./(1 + 25*x.^2);
xx = linspace(-1, 1, 20001);                    % fine grid
for n = [5 10 20]
  xe = linspace(-1, 1, n+1);                    % equally spaced
  xc = cos(pi*(2*(0:n) + 1)/(2*n + 2));         % Chebychev
  pe = polyfit(xe, f(xe), n);  pc = polyfit(xc, f(xc), n);
  ee = max(abs(f(xx) - polyval(pe, xx)));
  ec = max(abs(f(xx) - polyval(pc, xx)));
  fprintf('%3d  %9.3g  %9.3g\n', n, ee, ec)
end
```

- `polyfit` uses the monomial basis: for $n = 20$ (equally spaced) its matrix has $\kappa_2 \approx 8.3 \cdot 10^8$ (Lecture 17). The errors on the next slide are still correct to the digits shown.
- The maximum over a grid is a lower bound for the true maximum; 20001 points are enough here.

Runge's Example Explained

Example. $f(x) = 1/(1+25x^2)$ on $[-1,1]$; maximum error (computed):

n	5	10	20
equally spaced	0.43	1.92	59.8
Chebychev	0.56	0.11	0.015

- M is large: $f = \sum_{m\geqslant 0}(-25x^2)^m$ for $|x| < 1/5$, so $\left|f^{(m)}(0)\right|/m! = 5^m$ for even m; hence $M/(n+1)! \geqslant 5^{n+1}$ for odd n.
- Equally spaced: $|\omega|$ is large near the ends; the error there grows with n.
- Chebychev: the bound of the corollary is $\geqslant 5^{n+1}2^{-n}$ for odd n and grows, yet the observed error decreases. A bound is only an upper bound. (Fact, proof omitted: for every Lipschitz continuous f, the Chebychev interpolants converge to f uniformly on $[-1,1]$.)
- Rule: high degree $\Rightarrow$ Chebychev nodes; fixed grid $\Rightarrow$ splines.

Error of a Degree 1 Spline

Corollary. Let f'' be continuous on $[a, b]$, S the degree 1 spline with $S(t_i) = f(t_i)$, $i = 0, \ldots, n$, and $h := \max_i h_i$. Then

$$\max_{x \in [a,b]} |f(x) - S(x)| \leqslant \frac{h^2}{8} \max_{[a,b]} |f''| \qquad (O(h^2)\text{: halving } h \text{ divides the bound by } 4).$$

Proof. 1. One interval. For $x \in [t_i, t_{i+1}]$, S is the line through f at t_i, t_{i+1}. Interpolation Error I with $n = 1$ on $[t_i, t_{i+1}]$: for some $c \in (t_i, t_{i+1})$,

$$f(x) - S(x) = \frac{f''(c)}{2} (x - t_i)(x - t_{i+1}).$$

2. The product is largest at the midpoint: $|(x - t_i)(x - t_{i+1})| \leqslant (h_i/2)^2 \leqslant h^2/4$.

3. Combine: $|f(x) - S(x)| \leqslant \frac{1}{2} \cdot \frac{h^2}{4} \max_{[a,b]} |f''|$, for every x in every interval. ■

Error of a Cubic Spline

Theorem (*Hall–Meyer 1976; proof omitted*). Let $f^{(4)}$ be continuous on $[a,b]$, and S the fixed-slope cubic spline with $S(t_i) = f(t_i)$ and end slopes $f'(t_0)$, $f'(t_n)$. With $h := \max_i h_i$,

$$\max_{[a,b]} |f - S| \leqslant \frac{5}{384} h^4 \max_{[a,b]} \left| f^{(4)} \right| .$$

Example. $f = \sin$ on $[0, \pi/2]$, n equal intervals; maximum error (computed):

n	4	8	16	32	ratio
degree 1	$1.9 \cdot 10^{-2}$	$4.8 \cdot 10^{-3}$	$1.2 \cdot 10^{-3}$	$3.0 \cdot 10^{-4}$	≈ 4
natural	$7.7 \cdot 10^{-3}$	$1.9 \cdot 10^{-3}$	$4.7 \cdot 10^{-4}$	$1.2 \cdot 10^{-4}$	≈ 4
not-a-knot	$6.0 \cdot 10^{-4}$	$4.1 \cdot 10^{-5}$	$2.6 \cdot 10^{-6}$	$1.6 \cdot 10^{-7}$	≈ 16
fixed-slope	$6.3 \cdot 10^{-5}$	$3.9 \cdot 10^{-6}$	$2.4 \cdot 10^{-7}$	$1.5 \cdot 10^{-8}$	≈ 16

(Check, $n = 4$: $h^2/8 = 0.019$ and $\frac{5}{384} h^4 = 3.1 \cdot 10^{-4}$.)

Natural: only $O(h^2)$ here: it forces $S''(\pi/2) = 0$, but $\sin''(\pi/2) = -1$.

Measuring the Order in Matlab

If $E(h) \approx Ch^p$, halving h gives $E(h)/E(h/2) \approx 2^p$, so $p \approx \log_2\big(E(h)/E(h/2)\big)$.

```
a = 0;  b = pi/2;  xx = linspace(a, b, 20001);
ns = [4 8 16 32];  err = zeros(size(ns));
for m = 1:numel(ns)
  t = linspace(a, b, ns(m) + 1);
  S = spline(t, [cos(a) sin(t) cos(b)]);  % end slopes f'=cos
  err(m) = max(abs(sin(xx) - ppval(S, xx)));
end
p = log2(err(1:end-1)./err(2:end))          % observed order
```

Observed p for the table on the previous slide:

	degree 1	natural	not-a-knot	fixed-slope
p	2.0	2.0	3.9–4.0	4.0

(Not-a-knot: `spline(t, sin(t))`; natural: `natspline`.) Not-a-knot needs no end slopes and still gives order 4: this is why it is the default in `spline(t, y)`.

From Degree to Order

Given: $\Delta = \{x_0, x_1, \ldots, x_n\}$ with $a = x_0 < x_1 < \cdots < x_n = b$, $n \geqslant 1$. (These are the knots t_i of Lectures 17–19.)

Definition (*Spline of order k*). Let $k \geqslant 2$. S is a *spline of order k* on Δ if $S \in C^{k-2}[a, b]$ ($S, S', \ldots, S^{(k-2)}$ continuous) and, for $j = 1, \ldots, n$, the restriction of S to $[x_{j-1}, x_j]$ is in $\prod_{k-1}$ (polynomials of degree $\leqslant k-1$). $\mathcal{S}_{k,\Delta}$ denotes the set of all such splines.

Order k = degree $k-1$, the name used in Lecture 17 (each piece has k coefficients):

order k	degree $k-1$	name (Lectures 17–19)
2	1	degree 1 (linear) spline, C^0
3	2	degree 2 (quadratic) spline, C^1
4	3	cubic spline, C^2

Remark. $\mathcal{S}_{k,\Delta}$ is a real vector space: sums and real multiples of splines of order k are splines of order k. Also $\prod_{k-1} \subset \mathcal{S}_{k,\Delta}$.

The Dimension of $\mathcal{S}_{k,\Delta}$

Proposition. $\dim \mathcal{S}_{k,\Delta} = k + n - 1$.

Proof in Lecture 21 ("The Truncated Power Basis"): a basis with $k + n - 1$ elements. First, why this number:

Remark (*Count, as in Lecture 17*).

- On $[x_0, x_1]$: a polynomial of degree $\leqslant k - 1$, so k free coefficients.
- On each next interval: k coefficients, minus $k - 1$ matching conditions ($S, S', \ldots, S^{(k-2)}$ continuous at the knot): 1 new free coefficient.
- Total: $k + (n - 1)$.

The count is not a proof: it assumes the matching conditions are independent.

Remark (*Link to Lecture 18*). Interpolation gives $n + 1$ conditions, so $k + n - 1 - (n + 1) = k - 2$ conditions remain free. Cubic ($k = 4$): the 2 end conditions.

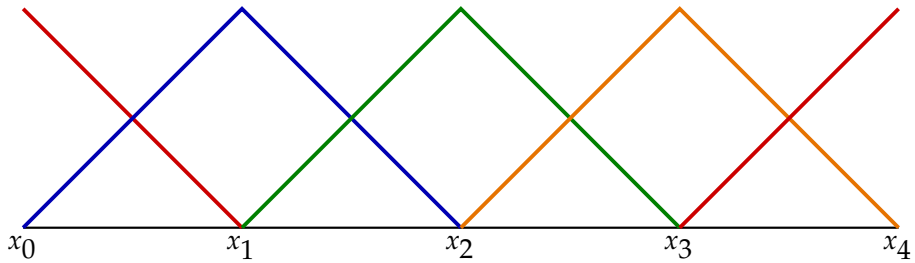
A Better Basis: Wish List

Wish list for a basis of $\mathcal{S}_{k,\Delta}$:

- *local:* each basis function is nonzero on at most k consecutive intervals;
- *nonnegative*, and *summing to* 1 at every $x \in [a, b]$;
- *well conditioned*, with cheap and stable evaluation.

The *B-splines* (defined below) have all three: the first two are proved, the third is measured, in Lecture 21.

Example ($k = 2$: *the hat functions*). On $x_0, \ldots, x_4$: N_{i2} is 1 at x_{i-1}, 0 at the other knots, linear in between (precise definition: two slides on):



$$S(x) = \sum_{i=1}^{n+1} y_{i-1} N_{i2}(x) \quad \text{satisfies} \quad S(x_{i-1}) = y_{i-1},\ i = 1, \ldots, n+1.$$

The coefficients are the data: no linear system at all.

Knots for B-Splines

B-splines are built on a *knot sequence*, in which a value may repeat:

$$\tau_1 \leqslant \tau_2 \leqslant \cdots \leqslant \tau_l, \qquad \tau_1, \tau_2, \ldots, \tau_l \in \mathbb{R}. \tag{1}$$

Definition (*Multiplicity*). The *multiplicity* of a knot value is the number of times it appears in (1).

For $\mathcal{S}_{k,\Delta}$: interior knots once, end points k times ("clamped"):

$$\overbrace{\underbrace{x_0}_{\tau_1=\cdots=\tau_k}}^{k \text{ times}} < \underbrace{x_1}_{\tau_{k+1}} < \cdots < \underbrace{x_{n-1}}_{\tau_{k+n-1}} < \overbrace{\underbrace{x_n}_{\tau_{k+n}=\cdots=\tau_{2k+n-1}}}^{k \text{ times}}$$

$$l = 2k + n - 1, \qquad l - k = k + n - 1 = \dim \mathcal{S}_{k,\Delta}.$$

Example. $k = 2$, $n = 4$: $\tau = (x_0, x_0, x_1, x_2, x_3, x_4, x_4)$, $l = 7$, $l - k = 5$ hat functions. The doubled end points produce the two half hats.

B-Splines: Definition

Definition (*B-splines; Cox–de Boor recursion*). The B-splines N_{ij} (index i, order j) of the knots (1) are defined by recursion on j (each order blends two neighbours of the order below, with weights linear in x):

$$N_{i1}(x) := \begin{cases} 1, & \tau_i \leqslant x < \tau_{i+1}, \\ 0, & \text{otherwise}, \end{cases} \qquad i = 1, 2, \ldots, l-1,$$

and for $j = 2, 3, \ldots, k$:

$$N_{ij}(x) := \frac{x - \tau_i}{\tau_{i+j-1} - \tau_i} N_{i,j-1}(x) + \frac{\tau_{i+j} - x}{\tau_{i+j} - \tau_{i+1}} N_{i+1,j-1}(x), \qquad i = 1, 2, \ldots, l-j. \quad (2)$$

Conventions:

1. A term whose denominator is 0 is set to 0 ("$0/0 = 0$"). Nothing is lost: e.g. $\tau_i = \tau_{i+j-1}$ makes $N_{i,j-1} \equiv 0$ (property (a), Lecture 21).
2. $N_{ik}(\tau_l) := \lim_{x \to \tau_l^-} N_{ik}(x)$, so that the right end point $b = \tau_l$ is covered.

Orders 1 and 2 by Hand

Take equally spaced knots $\tau_i = i$ (a long enough finite piece; no repeated knots). Order 1: boxes,

$$N_{i1}(x) = 1 \text{ for } i \leqslant x < i+1, \quad 0 \text{ otherwise.}$$

Order 2: (2) with $j = 2$, both denominators $= 1$:

$$N_{i2}(x) = (x-i)\,N_{i1}(x) + (i+2-x)\,N_{i+1,1}(x) = \begin{cases} x-i, & i \leqslant x < i+1, \\ i+2-x, & i+1 \leqslant x < i+2, \\ 0, & \text{otherwise.} \end{cases}$$

The hat function: up-ramp times the left box, plus down-ramp times the right box.

Remark. The weight $\frac{x-\tau_i}{\tau_{i+j-1}-\tau_i}$ rises linearly from 0 to 1 across $[\tau_i, \tau_{i+j-1}]$, where $N_{i,j-1}$ can be nonzero; the other weight falls from 1 to 0 across $[\tau_{i+1}, \tau_{i+j}]$. Each order widens the support (the set where the function is nonzero) by one interval.

Order 3 by Hand

Same knots, $j = 3$, both denominators $= 2$. With $s = x - i$:

$$N_{i3}(x) = \tfrac{s}{2} N_{i2}(x) + \tfrac{3-s}{2} N_{i+1,2}(x) = \begin{cases} \frac{1}{2}s^2, & 0 \leqslant s < 1, \\ \frac{1}{2}(-2s^2 + 6s - 3), & 1 \leqslant s < 2, \\ \frac{1}{2}(3-s)^2, & 2 \leqslant s < 3, \\ 0, & \text{otherwise.} \end{cases}$$

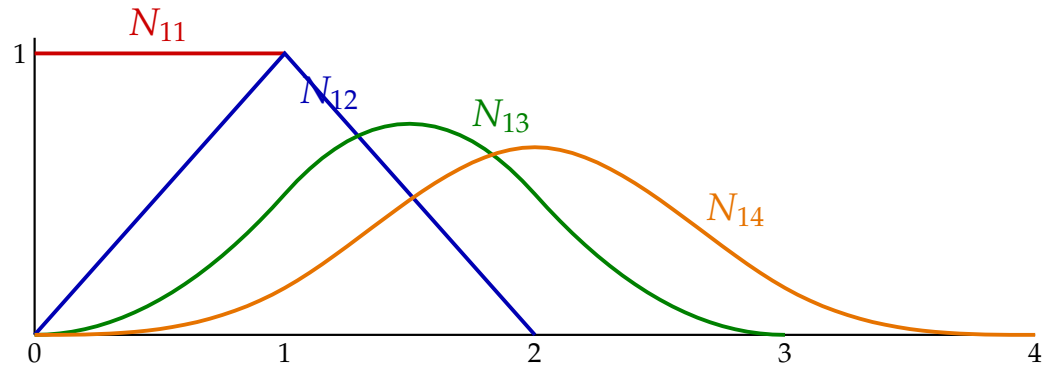
(Middle piece: $\frac{s}{2}(2-s) + \frac{3-s}{2}(s-1)$.)

- C^1 at $s = 1$: values $\frac{1}{2} = \frac{1}{2}$, slopes $1 = 1$; same at $s = 2$ by symmetry ($s \mapsto 3 - s$).
- At $x = 1.5$ only $N_{-1,3}, N_{0,3}, N_{1,3}$ are nonzero ($s = 2.5, 1.5, 0.5$):

$$\tfrac{1}{8} + \tfrac{3}{4} + \tfrac{1}{8} = 1.$$

Orders 1 to 4 on Equally Spaced Knots

$N_{11}, N_{12}, N_{13}, N_{14}$ for $\tau_i = i$: supports $[0,1]$, $[0,2]$, $[0,3]$, $[0,4]$.



- Each order: one more interval of support, one more continuous derivative, lower and wider.
- N_{14} is the cubic B-spline: C^2, and at the ends of its support $N_{14} = N'_{14} = N''_{14} = 0$.

Looking Ahead

- **Next lecture:** B-Splines: Properties, Evaluation, Interpolation